

PCB FAB

Complete Build & Process Documentation

PCB FAB

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Process

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Notes

1. Milling

Parts

18K+rpm spindle — permanently mounted to gantry Z carriage (not part of ATC)

Mounting bracket, fixed XY offset from ATC coupling point — aluminium plate, bolted to Z carriage alongside ATC coupling Motion system, 3 axis

Y = 200mm min

X = 200mm min

Z = 50mm max

Z probe (see Section 8 — Auto-Leveling & Registration Probe) Pin registry

3 pin asymmetric triangle

Holes placed in panel waste/tab material, not in the finished board outline

Can route or snap out depending on liability

Vacuum bed

Flipper

Process

1. Set blank down
2. Mill first side
3. Mill out registration pin holes — location calculated by software for optimal placement in waste/tab material
4. Machine inserts pins
5. Flip board
6. Set board down
7. Probe registers XY (cross-check vs pins) and Z mesh (see Section 8)
8. Mill out second side

2. Cleaning

Parts

Brushing - Nylon brush - DC motor, 1000–3000 RPM

Cleaning - Tub - IPA - 12V peristaltic pump - Tubing (angled right ahead of the brush)

Drying - 12V centrifugal fan (any) - Flat nozzle, angled behind the brush (barely an angle, right behind brush) - 12V centrifugal fan - Fine point nozzle, aimed at 45 degrees down

General - Drip pan below

Process

1. Clean first side with brush, drip, and flat dryer; fine tip dryer active — 3 passes
2. Go to vias and activate fine point nozzle
3. Flip and repeat
4. Wait 15 minutes

3. Solder Mask / Silk Screen

Parts

405nm 200mW+ UV laser (Civil Laser 300mW 405nm module) Slot die coater — 3D printable slot nozzle, fed by 12V peristaltic pump, sized for 150x150mm active board area

Liquid photoimageable solder mask (405nm curable)

12V peristaltic pump

Spray nozzle

Warm water

Small 12V heating element

UV flood/sun lamp

12V blower

UV silk screen ink (white, direct cure preferred)

Process

1. Lay down solder mask via slot die coater
2. Initial flood lamp for X seconds — tack cure to harden slightly
3. Laser direct write exposure (405nm)
4. Mix sodium carbonate with warm water, spray at uncured areas — develop
5. Water rinse spray — stops development
6. Air blast dry — gantry blower, gentle warm air
7. Final UV lamp cure — full harden
8. Put down silk screen ink via slot die coater (flush/swap fluid first)
9. Cure with UV laser (or flood cure if using direct-cure ink — no develop needed)
10. Flip
11. Probe registers XY/Z (see Section 8)

12. Repeat for Side B

4. Soldering

Parts — Solder Dispensing Head (Paste Auger)

1x NEMA 17 stepper motor
PETG filament — auger body and screw
1x 5mm shaft coupler
1x M3 grub screw
Pack of blunt luer lock dispensing needles (22 gauge)
1x 60–100ml luer lock syringe — reservoir
300–500mm, 3–4mm ID silicone tubing
4 wire stepper cable
Barb fittings and luer lock adapters

Parts — Hot Air System

Micro nozzle, 2–3mm ID
Large nozzle, 6–10mm ID
24V centrifugal blower, 10–15W (5–25W acceptable)
24V cartridge heater, 150–200W (120W min, 300W max)
3x K-type thermocouples (heater block + each nozzle)
Needle valve (optional, fine airflow control)
T-splitter (dual-nozzle setup)
Silicone or PTFE tubing, 6–8mm ID, 8–10mm OD
Aluminum or brass block, 20x20x40mm
4–5mm drill bit (air channel)
M6 or 1/8" NPT tap (fittings)
2x barbed hose fittings, 6–8mm barb
6mm drill bit (heater hole)
Optional M3/M4 set screw + tap (lock heater)
K-type thermocouple bead (block)
2mm drill bit (thermocouple hole)
High-temp silicone tubing, 6mm ID (nozzle connections)
2x hose clamps
Optional: ceramic insulation
Optional: Kapton tape
Optional: mounting bracket

Process — Paste Dispensing

1. Gantry picks up paste auger head (ATC)
2. Probe Z touch-off confirms head tip position
3. For each pad in BOM, gantry moves to pad XY, stepper drives auger forward a calibrated step count, brief reverse (suck-back), Z lift, move to next pad

Process — Pick and Place

(See Section 6 — full PnP process)

Process — Hot Air Reflow

1. Power-up and stabilization

- Turn on 24V supply, heater, blower
- Set heater block target to 300–320°C
- Wait until heater block thermocouple is stable ($\pm 3^\circ\text{C}$)
- Verify nozzle thermocouples read within $\sim 20\text{--}40^\circ\text{C}$ of block

2. Side A — bulk preheat (large nozzle)

- Mount large nozzle (6–10mm ID)
- Blower PWM medium (40–60%)
- Hold nozzle 20–40mm above PCB
- Slow overlapping passes over populated area
- Target $\sim 120\text{--}160^\circ\text{C}$ surface temp
- 30–90 seconds depending on board thickness/copper
- Do not exceed $\sim 180^\circ\text{C}$ or allow discoloration

3. Side A — focused reflow (micro nozzle)

- Switch to micro nozzle (2–3mm ID)
- Blower PWM low (15–30%)
- Heater block 300–330°C
- Hold nozzle 5–15mm above target
- Gentle circular/zig-zag motion per component group
- Watch paste: matte/gray $\rightarrow$ glossy $\rightarrow$ flowed/rounded
- Move on once joints are shiny with formed fillets
- Larger components: slightly increase PWM/dwell, keep motion continuous

4. Side A — cool-down

- Move nozzle away
- Optionally reduce heater power/airflow briefly
- Cool until solder is visibly solid (30–90 seconds)
- Do not flex/move board

5. Flip for Side B

- Flip PCB, Side A components supported on flat non compressive surface (fixture/rails/cutout)
- Confirm no tall Side A components bearing load or pushed sideways

Probe registers XY/Z (see Section 8)

6. Side B — bulk preheat (large nozzle)

- Same as Side A bulk preheat
- Target $\sim 120\text{--}160^\circ\text{C}$ on Side B while ensuring Side A doesn't reflow

7. Side B — focused reflow (micro nozzle)

Same technique as Side A

Periodically check Side A isn't overheating/shifting

8. Handling large thermal masses (both sides)

Extra preheat with large nozzle over wider area

Micro nozzle closer (5–10mm), slightly higher PWM,

300–330°C Approach from side, “sneak up” on joint

Stop once solder wets and forms smooth fillet

9. Final cool-down

Move nozzle away, reduce heater power

Cool naturally in still air — no direct cold air (thermal shock) Wait until all joints fully solid

10. Touch-up soldering (if needed)

Micro nozzle, low PWM, 280–320°C

5–10mm away, short targeted passes

Stop immediately once joint reflows/smooths

5. Pick and Place — Parts

Z Axis

T8 lead screw, 8mm, 2–4mm pitch, 150–200mm

length T8 brass lead screw nut

Flexible shaft coupler, NEMA 17 to 8mm

MGN12 linear rail with MGN12H carriage, 150–200mm

NEMA 17 stepper motor, standard torque, 1.8°

Z-axis mounting plate (aluminum)

Z-axis top and bottom brackets

Mechanical endstop switch (Z home)

Optional upper limit switch

Single-point load cell, 3kg, 2mV/V — placement force
feedback HX711 load cell amplifier module

Load cell mounting bracket

Shielded 4-wire cable

Small cable chain / wire loom for Z

High-flex wires for Z motor and sensors

Head

NEMA 11 stepper motor (rotation)

3–4mm steel rotation shaft

2x radial bearings (e.g. 4x8x3 or 5x10x4mm)

Small shaft hub/coupler (motor to shaft)

Nozzle holder (PnP-style or 3D-printed clamp)

Vacuum nozzles: 0.5mm, 1mm, 2mm, 3mm, plus flat/SQN

tip Spring-loaded nozzle mount, 1–2mm travel
Light compression spring
Rotation home sensor (mechanical or Hall) +
magnet/flag Small diaphragm vacuum pump, 3–8 L/min,
12V or 24V DIY vacuum reservoir (small bottle with
fittings)
Compact vacuum solenoid valve, 12V or 24V
Inline air filter
Vacuum tubing, 3–4mm ID silicone or PU
Hose barbs and fittings
USB 720p/1080p board camera, M12 lens mount
M12 lens, ~6mm focal length
Rigid camera bracket (frame-mounted)
USB cable, strain-relieved
12V LED ring light, 30–50mm inner diameter
5000–6500K white LEDs
Simple 12V LED driver or resistor-limited supply
Small N-MOSFET + resistor (optional PWM
dimming) Diffuser (frosted plastic or tape)
LED ring mounting bracket
Head structural plates (3D-printed or aluminum)
Standoffs and spacers

6. Pick and Place — Belt & Tray Feeder

Parts

GT2 belt 6mm (~1.2m loop)
GT2 pulley 20T, 5mm bore
Idler pulley smooth, 5mm bore GT2
NEMA 17 stepper, shaft coupler 5mm-5mm flex
Belt tensioner (printed)
MGN9 linear rail, 400mm x2
MGN9H carriage x4
Side guide channels (printed) x2
Vertical magazine frame
Release solenoid, 12V push
Solenoid mount (printed)
Magazine side guides (printed) x4
Compression spring ~5N
Optical sensor (slot/reflective)
Hard stop (printed)
Position encoder strip
Encoder reader module
Exit ramp (printed)
Collection bin
Stepper driver (TMC2209/A4988)

Solenoid driver (MOSFET module)
JST connectors 4-pin x4
Limit switch
M3x8 SHCS x30
M3x12 SHCS x15
M3 heat set inserts x30
M5x10 BHCS x12
M5 T-nuts x12
Motor mount (printed)
Sensor mounts (printed) x2

Trays

200x 0402 trays
150x 0603 trays
150x 0805 trays
150x 1206 trays
150x passive trays
3x per common IC
2x per misc component

Tray pocket dimensions, depths, chamfers, and materials per package — see SMD Tray Specifications reference (smd_tray_specs_v4.csv).

6a. Pick and Place — Process

Tray Feed

1. Pre-loaded trays stacked in hopper magazine, one component per pocket, in BOM placement order
2. Solenoid drops bottom tray onto belt
3. Belt advances tray to pickup zone — hard stop + optical sensor confirm position
4. Encoder tracks belt position — software knows which component is staged

Pickup

1. Gantry moves PnP head to pickup zone XY
2. Z lowers vacuum needle to component in tray pocket
3. Vacuum engages, component lifts
4. Z raises

Bottom Vision Check

1. Gantry moves component over bottom camera

2. Camera captures orientation and offset from needle center
3. Software calculates rotation and XY correction needed

Placement

1. Gantry moves to target pad XY on PCB, pre-corrected for offset
2. Rotation axis adjusts component orientation per vision data
3. Z lowers component into solder paste on pads
4. Vacuum releases
5. Z raises

Tray Cycle Continues

1. Belt advances — empty tray exits to collection bin
2. Hopper drops next tray
3. Repeat for next component in sequence

6b. Via Rivet Insertion Head

Parts

Small solenoid, 12V push type — internal press stroke
(self contained, not gantry Z force)
Spring return for solenoid plunger
Solenoid mount (printed PETG)
Hollow vacuum needle, 0.6–0.7mm ID (for 0.8–1mm pre-flanged
via rivets)
Single-point load cell, 1kg — monitors insertion
force HX711 load cell amplifier module
Vibratory bowl feeder, small, 12V — rivet supply and
orientation Pre-flanged via rivets, 0.8–1mm — bulk
Shared vacuum line from PnP vacuum system
ATC coupling plate (magnetic + key, per Section 7)

Why pre-flanged rivets and a self-contained press

Gantry Z does not bear insertion force — avoids stressing
the vacuum bed, Z leadscrew, or FR4 blank
Pre-flanged rivets (flanged on both ends) require no anvil/swage
from below — straight insertion seats both flanges by
interference fit
Load cell allows the head to detect misalignment or a bent
rivet before damage occurs

7. Auto Tool Head Changer

Parts

Gantry Coupling Plate - Neodymium disc magnets, 12mm x 5mm N52 x4 (lighter heads: 10x3mm N42) - Key pin (D-shaped/triangular, printed or machined) x1 - Alignment cone (printed PETG) x3 - Aluminium mounting plate, 3mm x1

Per-Head Plate (x6 — laser, cleaning, slot coater, paste auger, hot air, PnP) - Mild steel counter plate, 1–2mm - Key socket - Mounting plate (printed PETG)

Dock (x6) - Recessed dock pocket (printed PETG) - Retention spring clip x2 per dock - Dock frame mount, aluminium L-bracket

Overhead Routing (x6) - Service loop arm (printed PETG) - Flexible tubing 3–4mm ID (air/vacuum/fluid as needed) - Shielded signal cable (sensors/thermocouples/camera) - Power cable, 2-wire - Small cable clips (printed PETG) x2 per head

Rack Frame - 2020 extrusion, 500mm x1 - T-slot nuts M5 x12 - M5x10 BHCS x12

Hardware - M3x8 SHCS x30 - M3 heat set inserts

x30 Process

1. Gantry finishes current operation, moves to dock of currently mounted head
2. Lowers head into dock pocket — retention spring clips snap over the lip on the head's mounting plate
3. Gantry lifts straight up — magnetic holding force overcome by spring clip retention, head stays seated, gantry separates cleanly
4. Overhead service loop for that head flexes/retracts slightly — tubing and cables remain connected throughout, nothing disconnects
5. Gantry moves to dock of next required head
6. Lowers toward the head — alignment cones guide rough positioning
7. Key pin enters key socket — prevents rotation, finalizes XY/rotational alignment
8. Continued descent — magnets fully engage, head seats against gantry coupling plate
9. Gantry lifts — head lifts with it, spring clips release as head clears dock pocket
10. The overhead service loop extends with the head as it moves to work area
11. **Probe does a quick Z touch-off to confirm this**

head's tip position before starting work

12. Operation proceeds

8. Auto-Leveling & Registration Probe

Purpose

A single combined probe handles two functions on every flip and every tool change: - **XY edge-finding** — cross-check against registration pin position - **Z surface mapping** — builds a height mesh to compensate each tool head's Z offset, accounting for board warp and thickness changes between operations

Parts

- Single-point load cell, 1–3kg, 2mV/V
- HX711 load cell amplifier module
- Small dedicated Z axis for probe
 - NEMA 8 stepper or small linear actuator/solenoid
 - 5–10mm travel — enough to extend 0.2mm below the lowest point of any mounted tool head
 - Mini linear rail or guided sleeve for repeatable extend/retract
- Probe tip — rounded ball or Delrin tip, small diameter (avoids scratching surfaces/catching board edges)
- Fixed mounting bracket — known XY offset from gantry coupling point and mill spindle
- Shielded 4-wire cable (load cell signal)
- High-flex wiring for small Z axis

Process — Every Flip

1. Board flips onto registration pins — primary XY/orientation reference
2. Probe deploys (extends 0.2mm below current head's lowest point)
3. Probe contacts board edge on two adjacent sides — XY reading taken
4. Software compares probe XY reading to pin-expected position
 - Within tolerance (0.05–0.1mm): proceed
 - Outside tolerance: flag for inspection (possible pin wear/seating issue)
5. If XY check passes — probe runs Z mesh across board (grid of touch points)
6. Software builds Z compensation map for the upcoming operation
7. Probe retracts
8. Operation proceeds using pin registration (XY) + probe Z mesh (height compensation)

Process — Every Tool Change

1. After ATC pickup, probe deploys briefly
2. Quick single-point Z touch-off confirms new head's tip position
3. Probe retracts
4. Head's Z offset updated in software before operation begins

Frame

2020 Extrusion

Base perimeter X: 500mm x4
Base perimeter Y: 500mm x4
Vertical corner posts: 400mm x4
Top perimeter X: 500mm x4
Top perimeter Y: 500mm x4
Gantry X rail support: 350mm x2
Gantry Y rail support: 300mm x2
Belt system support rail: 300mm x2

Corner Hardware

3-way corner cube bracket x8
L-bracket 90 degree x24
T-slot nut M5 x100
M5x10 BHCS x100
M5x16 BHCS x20

Gantry Rails

MGN12 linear rail, 350mm x2 + MGN12H carriage x4
MGN12 linear rail, 300mm x2 + MGN12H carriage x4
GT2 belt 6mm ~1.5m x2
GT2 pulley 20T, 5mm bore x4
Idler pulley GT2 compatible x4
NEMA 17 stepper x2

Outer Panels

HDPE 5mm, 500x400mm x2 — wet process side panels
MDF 6mm, 500x400mm x2 — electronics/dry side panels
MDF 6mm, 500x400mm x1 — back panel
Acrylic 5mm, 500x400mm x1 — front access door
Acrylic or HDPE, 500x500mm x1 — top dust cover,
hinged HDPE 5mm, 500x500mm x1 — base panel

HDPE 5mm, 200x150mm x1 — drip tray

Heat Shield

Aluminium sheet 1mm, 200x200mm x1 — base/panel under reflow travel zone

Aluminium sheet 1mm, 150x100mm x1 — side panel near reflow travel

M3 standoffs 10mm x8 — heat shield mounting (air gap)

Panel Mounting

2020 panel mount clips x40

M3x10 self-tap x40

Hinges x2

Magnetic catches x2

Cable Management

Drag chain 10x10mm, 400mm x2

Drag chain mounts (printed) x4

Cable gland M12 x4

Ventilation

12V exhaust fan, 80mm x1

Fan grille, 80mm x1

Flexible ducting 50mm x500mm x1

General Hardware

M3x8 SHCS x50

M3x12 SHCS x30

M3 heat set inserts x50

M5x10 BHCS x100

M5 T-nuts x100 # PCB Fabrication — Full Process (FR4 Blank to Finished Board)

Bed Registration System Overview

The vacuum bed uses a motorized alignment pin carriage rather than fixed pins:

Three retractable alignment pins mounted on a motorized carriage (lead screw + guide rods + stepper/servo per pin or shared coordinated mechanism)

Pins individually retract below the bed/copper surface for milling, and extend up through tooling holes for PnP and flip registration. Controller stores pin carriage positions per board — once tooling holes are drilled, the carriage moves to match and that position is reused for all subsequent operations on that board.

Large support pins — separate retractable pins located under panel rails/keepout/bare FR4 zones, used only during second-side PnP to support the board when first-side components are on the underside. Never used during routing.

Operating Modes

PnP Mode - Alignment pins UP, inserted through tooling holes - Support pins UP only when components exist on the underside (second-side PnP) - Vacuum ON

Routing/Milling Mode - Alignment pins DOWN (fully retracted below copper surface) - Support pins DOWN - Vacuum ON - Probe builds Z height map, milling proceeds with compensation

Pins retract for milling to avoid fighting the vacuum, distorting probe Z readings, risking tool collisions, and introducing vibration paths.

Stage 1 — Blank Insertion & Tooling Hole Creation

Parts used: Vacuum bed, mill spindle (permanent mount), Z probe, motorized alignment pin carriage, FR4 blank

1. Place raw FR4 blank on vacuum bed (vacuum ON), alignment pins DOWN, support pins DOWN
2. Auto-locate the board — edge probing (auto-leveling probe) and/or camera outline detection
3. Probe deploys — Z mesh of blank surface
4. Mill Side A — traces, pads, through-holes per Gerber (Routing Mode)
5. Drill 3 tooling holes in safe zones (panel waste/tab material) — locations calculated by software
6. Motorized pin carriage moves — aligns all three pins to match the drilled tooling hole pattern
7. Alignment pins extend UP through the tooling holes — board now permanently registered for all subsequent operations
8. Probe retracts

Stage 1a — Via Rivet Insertion

Parts used: Via insertion head (internal solenoid press, load cell, hollow vacuum needle, vibratory bowl feeder, pre-flanged via rivets)

- 8a. ATC swaps to via insertion head
- 8b. Probe Z touch-off confirms insertion head tip position
- 8c. Vibratory bowl feeder supplies oriented

pre-flanged rivets to pickup point 8d. For each via location in Gerber data: - Gantry moves to via XY - Hollow vacuum needle picks one rivet from feeder pickup point - Z lowers rivet over the drilled via hole - Internal solenoid presses rivet into hole — pre-flanged ends grip both copper layers - Load cell monitors insertion force — stops/flags if resistance indicates misalignment or bent rivet - Vacuum releases, Z raises - Move to next via 8e. All vias inserted before cleaning begins — both flanges will be soldered during reflow (Stages 11/15)

Stage 2 — Side A Cleaning

Parts used: Cleaning head (nylon brush, DC motor, IPA peristaltic pump/tubing, flat + fine-point dryer nozzles, centrifugal fans), drip pan

9. ATC swaps to cleaning head

10. Probe Z touch-off confirms cleaning head tip position 11. Brush + IPA drip + flat dryer + fine-tip dryer active — 3 passes over Side A

12. Activate fine point nozzle over via locations

13. Wait 15 minutes (copper oxidation window)

Stage 3 — Side A Solder Mask & Silkscreen

Parts used: Slot die coater head, 12V peristaltic pump, LPSM, 405nm 300mW laser head, UV flood lamp, spray head, sodium carbonate, warm water, gantry blower, UV silkscreen ink

14. ATC swaps to slot die coater head

15. Probe Z touch-off

16. Lay down solder mask (LPSM) via slot die coater

17. ATC swaps to laser head (or same head if combined) 18. Initial flood lamp — tack cure (X seconds)

19. Laser direct write exposure (405nm) — mask layer from Gerber 20. ATC swaps to spray head

21. Sodium carbonate + warm water spray — develop uncured areas

22. Water rinse spray — stop development

23. Air blast dry (gantry blower, gentle warm air)

24. Final UV flood lamp cure — full harden

25. ATC swaps to slot die coater — flush lines, load silkscreen ink 26. Lay down silkscreen ink via slot die coater

27. ATC swaps to laser head

28. Laser direct write exposure — silkscreen layer

29. Flood UV cure (direct-cure ink — no develop needed)

Stage 4 — Flip to Side B (for Milling)

Parts used: Flipper (corner grippers + dual load cell alignment arm), motorized alignment pin carriage, support pins, auto-leveling probe

30. Alignment pins retract DOWN (Routing Mode prep), support pins remain DOWN — Side A has no components yet
31. Vacuum releases
32. Flipper arms extend from wall — corner gripper arms + lower load-cell alignment arm extend together
33. Corner vacuum cups engage at two board corners
34. Flipper assembly lifts — board clears bed surface
35. Single rotation motor rotates assembly 180 degrees
36. Flipper lowers — load cell alignment arm tip contacts bed surface first, signals Z landing height, descent stops
37. Horizontal load cell on alignment arm detects any XY shift — gantry/alignment nudges board until pins seat fully and lateral force reads neutral
38. Alignment pins extend UP through tooling holes — board re registered
39. Vacuum engages
40. Corner grippers release, flipper retracts to wall
41. Probe deploys — XY edge check vs pin-expected position (tolerance 0.05–0.1mm), then Z mesh of Side B
42. Probe retracts
43. Alignment pins retract DOWN for milling (Routing Mode)

Stage 5 — Side B Milling

Parts used: Mill spindle (permanent mount), Z probe

44. Mill Side B — traces, pads, through-holes per Gerber (Routing Mode, pins down, Side B Z mesh applied)
45. Do not drill new tooling holes — Side A holes/pin carriage position remain the reference
46. Alignment pins extend UP — return to PnP-ready state

Stage 6 — Side B Cleaning

Parts used: Cleaning head (nylon brush, DC motor, IPA peristaltic pump/tubing, flat + fine-point dryer nozzles, centrifugal fans), drip pan

47. ATC swaps to cleaning head
48. Probe Z touch-off

- 49. Brush + IPA drip + flat dryer + fine-tip dryer — 3 passes over Side B
- 50. Activate fine point nozzle over via locations
- 51. Wait 15 minutes

Stage 7 — Side B Solder Mask & Silkscreen

Parts used: Slot die coater head, 12V peristaltic pump, LPSM, 405nm 300mW laser head, UV flood lamp, spray head, sodium carbonate, warm water, gantry blower, UV silkscreen ink

- 52. ATC swaps to slot die coater
- 53. Probe Z touch-off
- 54. Lay down solder mask via slot die coater
- 55. ATC swaps to laser head
- 56. Initial flood lamp — tack cure
- 57. Laser direct write exposure — mask layer
- 58. ATC swaps to spray head
- 59. Sodium carbonate + warm water spray — develop 60. Water rinse spray
- 61. Air blast dry
- 62. Final UV flood cure
- 63. ATC swaps to slot die coater — flush, load silkscreen ink 64. Lay down silkscreen ink
- 65. ATC swaps to laser head
- 66. Laser direct write exposure — silkscreen layer
- 67. Flood UV cure

Stage 8 — Flip to Side A (for First-Side PnP)

Parts used: Flipper, motorized alignment pin carriage, support pins, auto-leveling probe

- 68. Alignment pins retract DOWN, support pins remain DOWN — neither side has components yet
- 69. Vacuum releases
- 70. Flipper extends, corner grippers engage at two corners 71. Assembly lifts, rotates 180 degrees
- 72. Lowers — load cell alignment arm signals Z landing, horizontal load cell confirms/corrects XY
- 73. Alignment pins extend UP through tooling holes
- 74. Vacuum engages
- 75. Flipper retracts

76. Probe deploys — XY edge check vs pins, Z mesh of Side A 77.
Probe retracts
78. PnP Mode active — alignment pins UP, support pins DOWN
(bottom is bare, board sits flat)

Stage 9 — Side A Paste Dispensing

Parts used: Paste auger dispenser head
79. ATC swaps to paste auger dispenser head
80. Probe Z touch-off confirms dispenser tip position, applies Side A Z mesh
81. For each pad in Side A BOM:
Gantry moves to pad XY
Auger stepper drives forward — calibrated step count for pad size
Auger reverses briefly (suck-back)
Z lifts
Move to next pad

Stage 10 — Side A Pick and Place

Parts used: PnP head (Z lead screw, NEMA 17, load cell, rotation NEMA 11, vacuum nozzles, vacuum pump/solenoid, bottom camera + LED ring), belt + tray feeder (hopper, solenoid, encoder, trays)
82. ATC swaps to PnP head
83. Probe Z touch-off confirms PnP head tip position, applies Side A Z mesh
84. Confirm PnP Mode — alignment pins UP, support pins DOWN, vacuum ON
Tray Feed 85. Pre-loaded trays stacked in hopper magazines, one component per pocket, in Side A BOM placement order 86. Solenoid drops bottom tray onto belt 87. Belt advances tray to pickup zone — hard stop + optical sensor confirms position 88. Encoder tracks belt position — software knows which component is staged
Pickup 89. Gantry moves PnP head to pickup zone XY 90. Z lowers vacuum needle to component in tray pocket 91. Vacuum engages, component lifts 92. Z raises
Bottom Vision Check 93. Gantry moves components over bottom camera 94. Camera captures orientation and offset from needle center 95. Software calculates rotation and XY correction needed
Placement 96. Gantry moves to target pad XY on PCB, pre-corrected for offset 97. Rotation axis adjusts component orientation per vision data 98. Z lowers components into solder paste on pads 99. Vacuum releases 100. Z raises

Tray Cycle 101. Belt advances — empty tray exits to collection bin 102. Hopper drops next tray 103. Repeat steps 89–102 for each component in Side A sequence

Stage 11 — Side A Reflow

Parts used: Hot air head (large + micro nozzles, 24V blower, 24V cartridge heater, K-type thermocouples)

104. ATC swaps to hot air head

105. Probe Z touch-off confirms hot air head tip position

Power-up and Stabilization 106. Turn on 24V supply, heater, blower

107. Set heater block target to 300–320°C 108. Wait until the heater blocks the thermocouple stable ($\pm 3^\circ\text{C}$) 109. Verify nozzle thermocouples within $\sim 20\text{--}40^\circ\text{C}$ of block

Bulk Preheat (Large Nozzle) 110. Mount/select large nozzle (6– 10mm ID) 111. Blower PWM medium (40–60%) 112. Hold nozzle 20– 40mm above PCB 113. Slow overlapping passes over populated area 114. Target $\sim 120\text{--}160^\circ\text{C}$ surface temp, 30–90 seconds 115. Do not exceed $\sim 180^\circ\text{C}$ or allow discoloration

Focused Reflow (Micro Nozzle) 116. Switch to micro nozzle (2– 3mm ID) 117. Blower PWM low (15–30%) 118. Heater block 300– 330°C 119. Hold nozzle 5–15mm above target 120. Gentle circular/zig-zag motion per component group 121. Watch paste: matte/gray → glossy → flowed/rounded 122. Move on once joints shiny with formed fillets 123. Larger components: slightly increase PWM/dwell, keep motion continuous

Cool-down 124. Move nozzle away 125. Optionally reduce heater power/airflow briefly 126. Cool until the solder is visibly solid (30–90 seconds) 127. Do not flex/move board

Stage 12 — Flip to Side B (for Second-Side PnP)

Parts used: Flipper, motorized alignment pin carriage, support pins, auto-leveling probe

128. Alignment pins retract DOWN

129. Vacuum releases

130. Flipper extends, corner grippers engage at two corners — grip points remain in tab/waste areas, clear of Side A components 131.

Assembly lifts — clearance confirmed for Side A components on the underside of the lift arc

132. Rotates 180 degrees

133. Lowers — load cell alignment arm signals Z landing (accounts for Side A component height under the board), horizontal load cell confirms/corrects XY
134. Alignment pins extend UP through tooling holes
135. Support pins extend UP — positioned under panel rails/keepout/bare FR4 zones to support the board around Side A components on the underside
136. Vacuum engages — stabilizes laterally
137. Flipper retracts
138. Probe deploys — XY edge check vs pins, Z mesh of Side B 139. Probe retracts
140. PnP Mode active — alignment pins UP, support pins UP (components on underside)

Stage 13 — Side B Paste Dispensing

Parts used: Paste auger dispenser head

141. ATC swaps to paste auger dispenser head
142. Probe Z touch-off, applies Side B Z mesh
143. For each pad in Side B BOM: dispense paste (same sub-process as Stage 9)

Stage 14 — Side B Pick and Place

Parts used: PnP head, belt + tray feeder

144. ATC swaps to PnP head
145. Probe Z touch-off, applies Side B Z mesh
146. Confirm PnP Mode — alignment pins UP, support pins UP, vacuum ON
147. Run full tray feed / pickup / vision / placement cycle (same sub process as Stage 10) for all Side B components

Stage 15 — Side B Reflow

Parts used: Hot air head (large + micro nozzles, 24V blower, 24V cartridge heater, K-type thermocouples)

148. ATC swaps to hot air head
149. Probe Z touch-off
Bulk Preheat (Large Nozzle) 150. Blower PWM medium (40–60%) 151. Hold nozzle 20–40mm above Side B 152. Overlapping passes over the entire soldering area 153. Target ~120–160°C on Side B while ensuring Side A does not reflow 154. 30–90 seconds, adjusted for board mass/copper

Focused Reflow (Micro Nozzle) 155. Switch to micro nozzle, blower PWM low (15–30%) 156. Heater block 300–330°C, nozzle 5–15mm above joints 157. Small circular/sweeping motions, nozzle always moving 158. Watch for matte → glossy → flowed with smooth fillets 159. Heavy copper/large pads: increase dwell/PWM slightly, avoid staying fixed 160. Periodically check Side A not overheating/shifting Handling Large Thermal Masses (Both Sides) 161. Extra preheat with a large nozzle over a wider area 162. Micro nozzle closer (5–10mm), slightly higher PWM, 300–330°C 163. Approach from side, “sneak up” on joint 164. Stop once solder wets and forms smooth fillet Final Cool-down 165. Move nozzle away, reduce heater power 166. Cool naturally in still air — no direct cold air (thermal shock) 167. Wait until all joints fully solid on both sides

Stage 16 — Touch-up (If Needed)

Parts used: Hot air head (micro nozzle)

- 168. ATC remains on hot air head
- 169. Micro nozzle, low PWM, 280–320°C
- 170. 5–10mm away, short targeted passes per joint
- 171. Stop immediately once joint reflows/smooths

Stage 17 — Completion

Parts used: Vacuum bed, motorized alignment pin carriage, support pins

- 172. Alignment pins retract DOWN, support pins retract DOWN 173. Vacuum off
- 174. Board removed from bed
- 175. Board separates from tooling-hole tabs at panel perimeter (if not already routed during Stage 1/5)
- 176. Through-hole components (manual) — insert and hand-solder separately, not part of automated sequence

Notes

Via rivets are inserted in Stage 1a immediately after Side A milling and tooling hole registration, before cleaning — both flanges are soldered automatically during reflow (Stages 11/15), no separate heating step required.

Tooling holes are drilled once during Stage 1, in panel waste/tab material. The motorized pin carriage repositions to match this pattern and the position is stored — reused for every subsequent flip on this board, never re-drilled.

Alignment pins retract for all milling operations (Stages 1, 5) and extend for all PnP and flip-registration operations.

Support pins extend only during Stage 12/14 (second-side PnP), positioned under panel rails/keepout/bare FR4 to protect Side A components on the underside. Retracted at all other times. The flipper alignment arm (dual load cell — vertical for Z landing, horizontal for XY correction) runs on every flip (Stages 4, 8, 12), working together with the auto-leveling probe's independent XY/Z verification pass.

Probe registration (Section 8 of build documentation) runs at every flip and after every ATC tool change — XY cross-checked against pins, Z mesh built fresh each time to account for warp/thickness changes from prior operations.

Tray BOM order must be generated separately for Side A and Side B components before Stage 10 / Stage 14 begin.

Tray pocket dimensions/materials — see SMD Tray Specifications reference (smd_tray_specs_v4.csv).